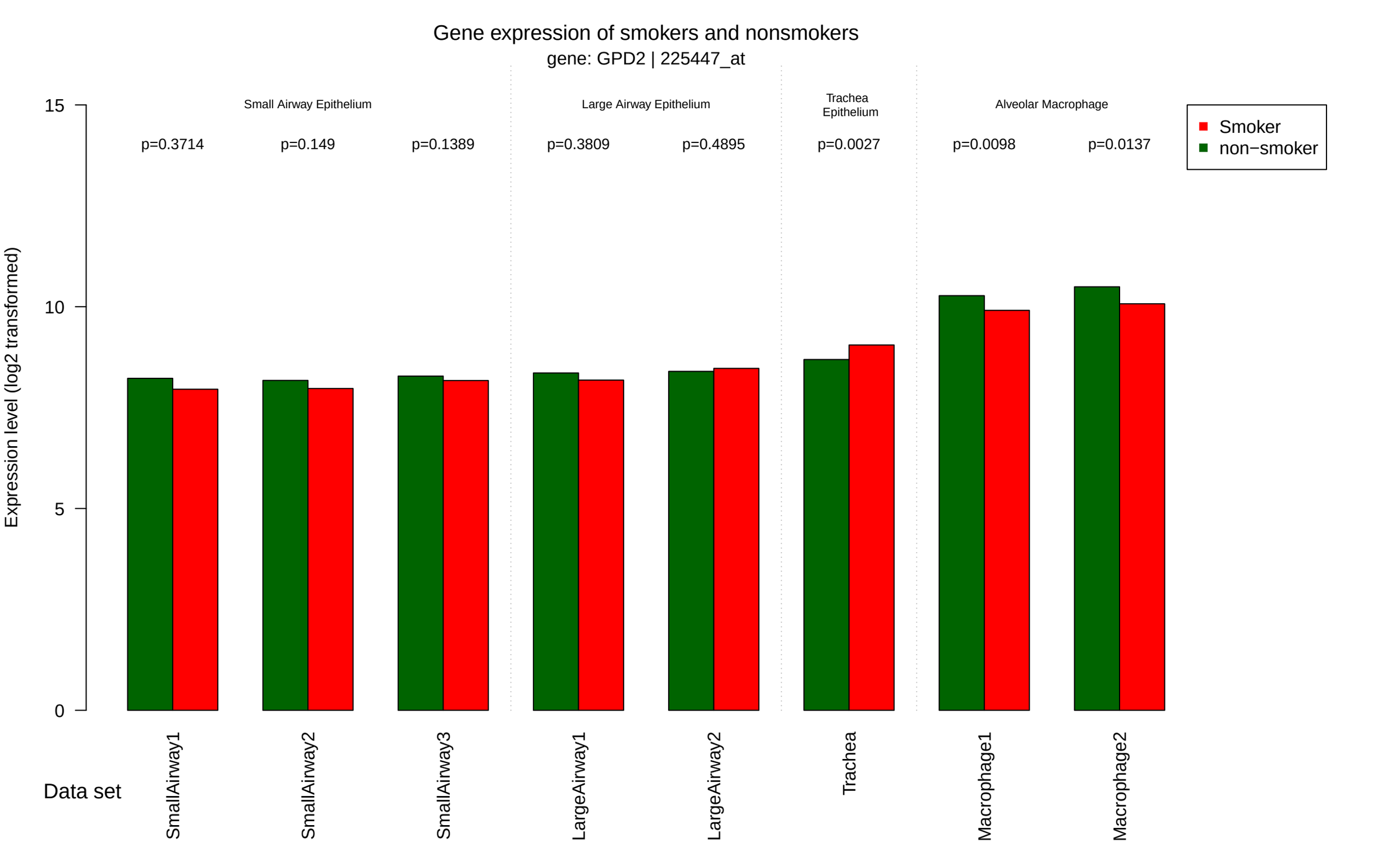
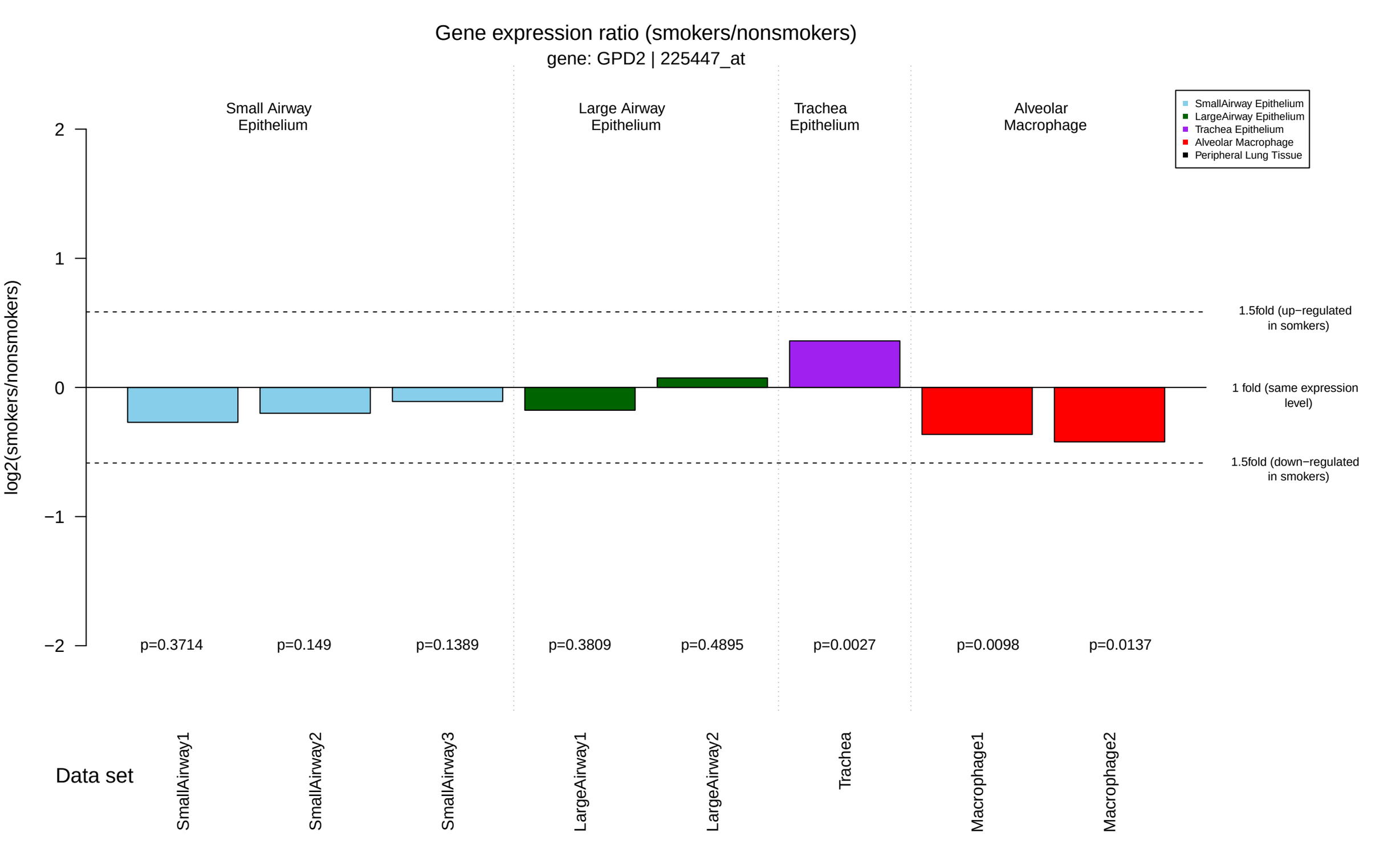
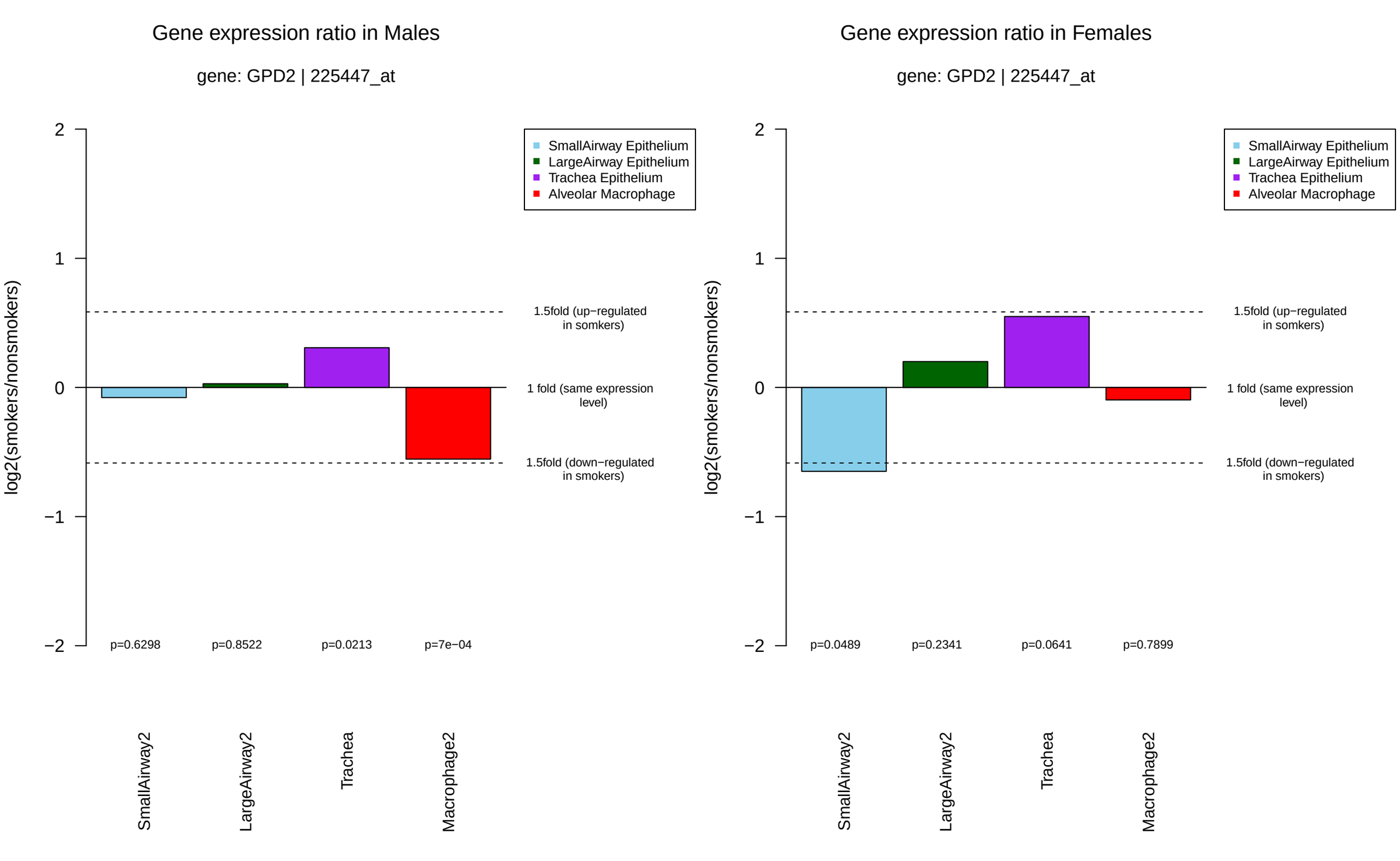
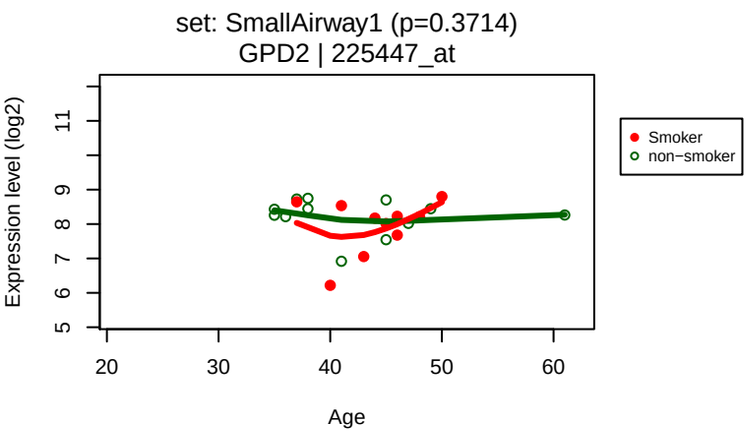
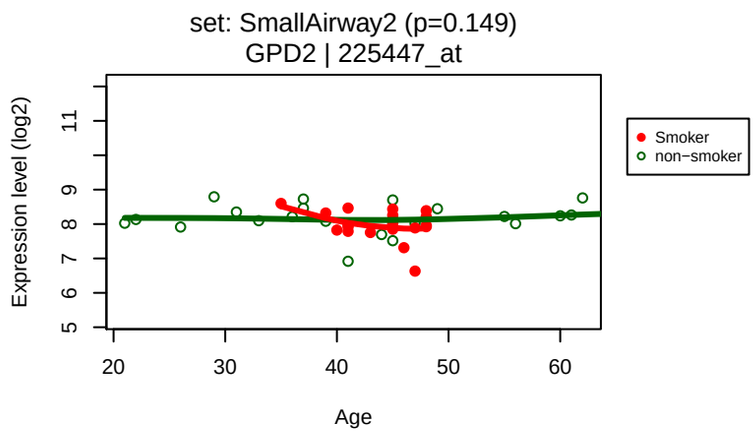
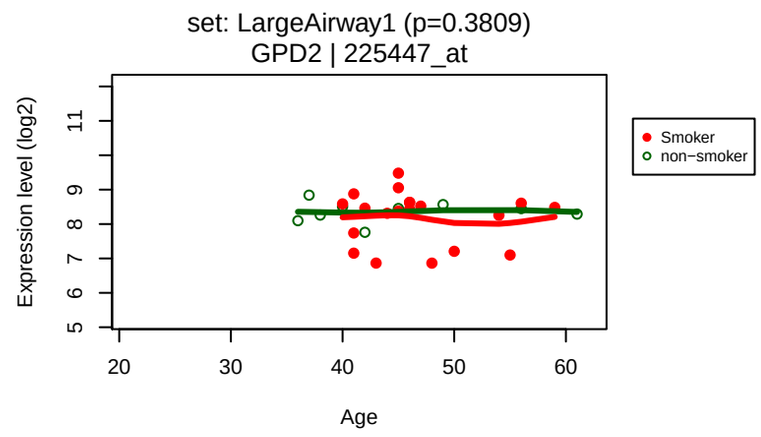

Gene expression of smokers and nonsmokers
gene: GPD2 | 225447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3714
p=0.149
p=0.1389
p=0.3809
p=0.4895
p=0.0027
p=0.0098
p=0.0137
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

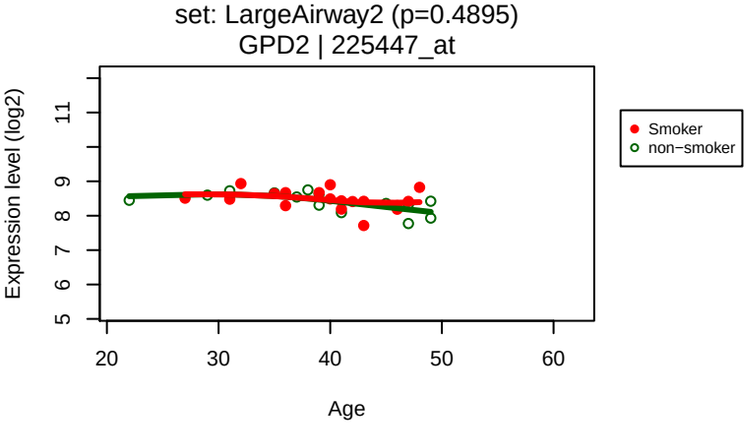

Gene expression ratio (smokers/nonsmokers)
gene: GPD2 | 225447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3714
p=0.149
p=0.1389
p=0.3809
p=0.4895
p=0.0027
p=0.0098
p=0.0137
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

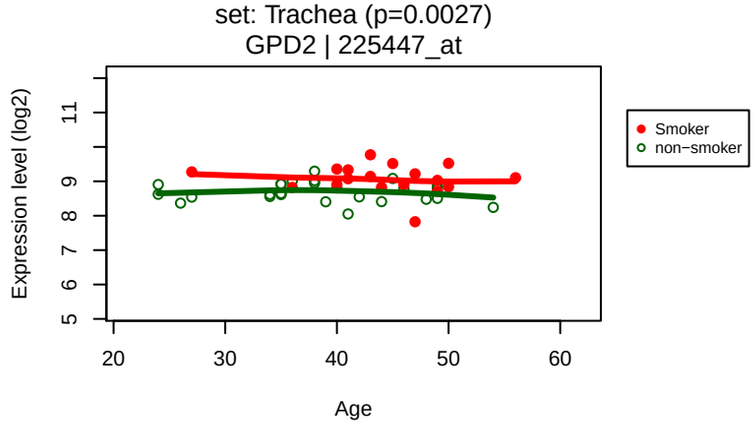

Gene expression ratio in Males
gene: GPD2 | 225447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6298
p=0.8522
p=0.0213
p=7e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPD2 | 225447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0489
p=0.2341
p=0.0641
p=0.7899
SmallAirway2
LargeAirway2
Trachea
Macrophage2

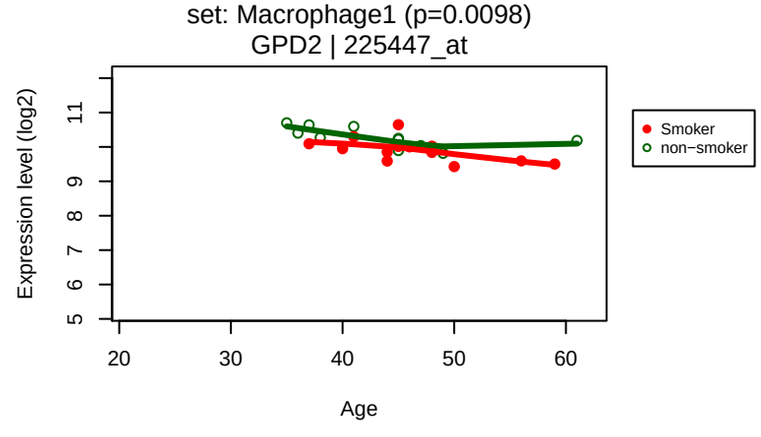

set: SmallAirway1 (p=0.3714)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker

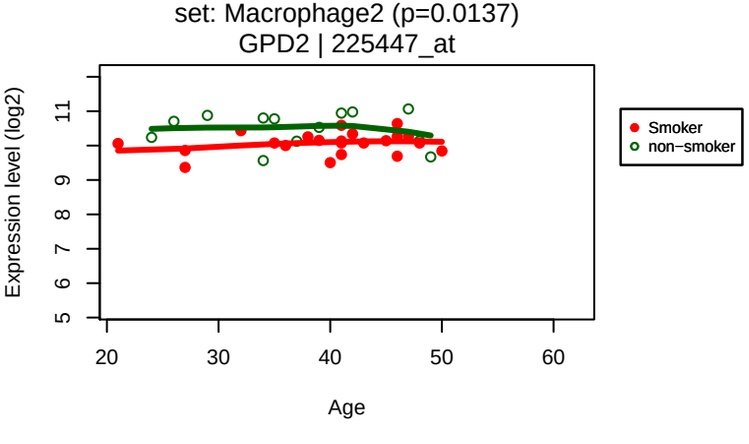

set: SmallAirway2 (p=0.149)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker

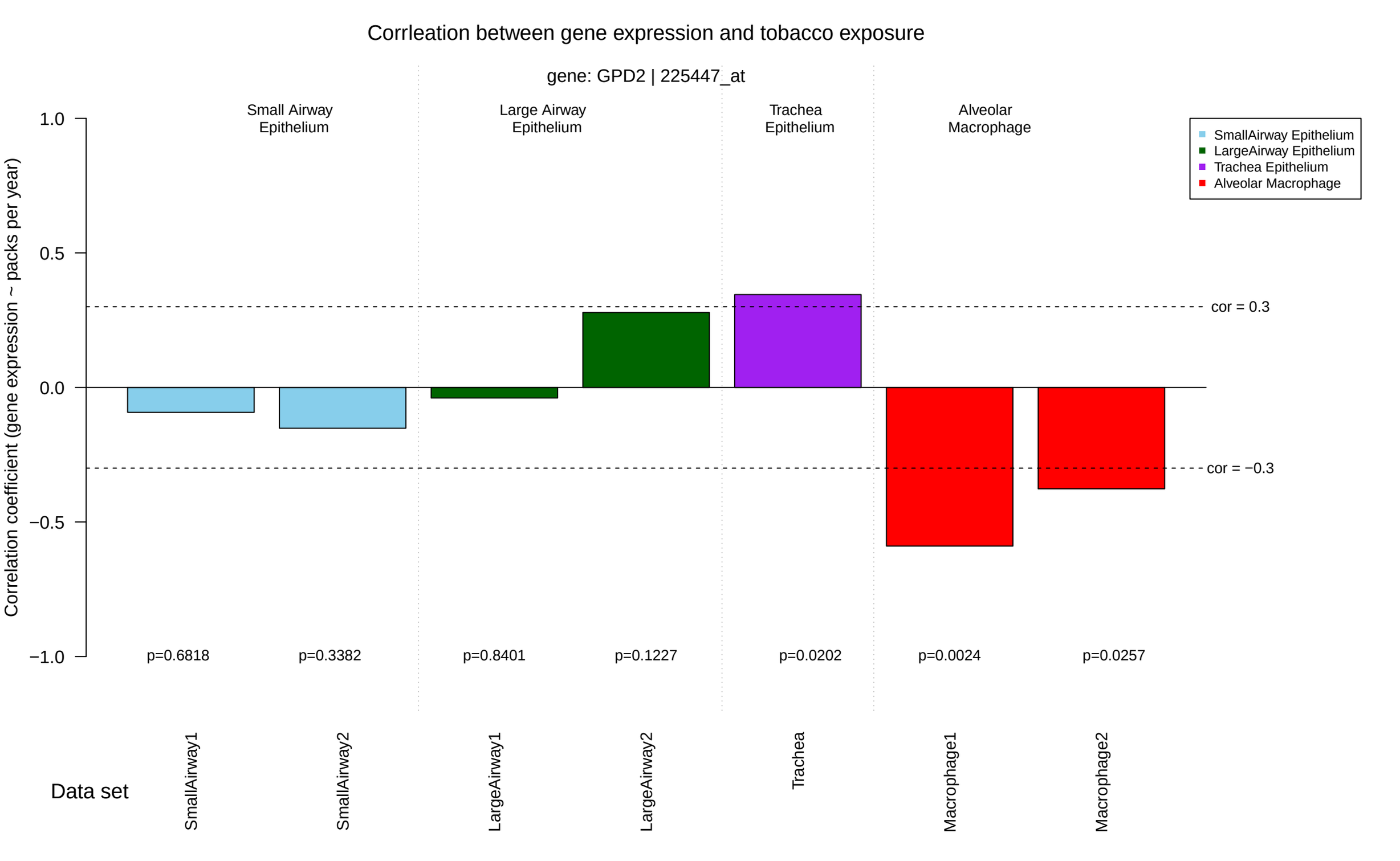

set: LargeAirway1 (p=0.3809)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4895)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0027)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0098)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0137)
GPD2 | 225447_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPD2 | 225447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6818
p=0.3382
p=0.8401
p=0.1227
p=0.0202
p=0.0024
p=0.0257
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2